Exercise of Supervised Learning: Information Theory Part 2

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May 28, 2026

Exercise 1: Entropy

A fair die is rolled at the same time as a fair coin is tossed. Let A be the number on the upper surface of the dice and let B describe the outcome of the coin toss, where

$$B = \begin{cases} 1, & \text{head,} \\ 0 & \text{tail.} \end{cases}$$

Two random variables X and Y are given by $X = A + B$ and $Y = A - B$, respectively.

(a) Calculate the entropies $H(X)$ and $H(Y)$, the conditional entropies $H(Y|X)$ and $H(X|Y)$, the joint entropy $H(X, Y)$ and the mutual information $I(X; Y)$.

Exercise 1 (a): Setup

$x = a + b \qquad a = \frac{x+y}{2}$

$y = a - b \qquad b = \frac{x-y}{2}$

Definitions

$$X = A + B, \qquad Y = A - B$$

Elementary probabilities

$$P(A = a) = \frac{1}{6}, \quad a = 1, \ldots, 6,$$

$$P(B = b) = \frac{1}{2}, \quad b = 0, 1,$$

$$P(A = a, B = b) = \frac{1}{12}.$$

Before $H(X)$ and $H(Y)$

Compute the joint distribution of X and Y by using the relation between (x, y) and (a, b).

One-to-one joint mapping

$$(a, b) \mapsto (x, y) = (a + b, a - b)$$

$$(x, y) \mapsto (a, b) = \left(\frac{x + y}{2}, \frac{x - y}{2}\right)$$

Consequence

$$P(X = x, Y = y) = P(A = a, B = b) = \frac{1}{12}$$

for every feasible (x, y).

Exercise 1 (a): Setup

$(a_1, b_1) \longleftrightarrow (x_1, y_1)$

$(a_2, b_2) \longleftrightarrow (x_2, y_2)$

Definitions

$$X = A + B, \qquad Y = A - B$$

Elementary probabilities

$$P(A = a) = \frac{1}{6}, \quad a = 1, \dots, 6,$$

$$P(B = b) = \frac{1}{2}, \quad b = 0, 1,$$

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Exercise 1 (a): Setup

Definitions

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Consequence

$$P(X = x, Y = y) = P(A = a, B = b) = \frac{1}{12}$$

for every feasible (x, y).

(A, B)	(X, Y)	$\mathbb{P}$
(a_1, b_1)	(x_1, y_0)	$1/12$
(a_2, b_2)	(x_2, y_1)	$1/12$

Exercise 1 (a): Joint Entropy

$P(X, Y)$

$H(X, Y)$

Joint entropy

$$H(X, Y) = -\sum_{x,y} p_{X,Y}(x, y) \log_2 p_{X,Y}(x, y)$$
$$= -12 \cdot \frac{1}{12} \log_2 \frac{1}{12}$$
$$= 2 + \log_2 3$$

Key fact

$$(a, b) \leftrightarrow (x, y)$$
$$p(x, y) = p(a, b) = \frac{1}{12}.$$

Exercise 1 (a): Joint Entropy

Joint entropy

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Key fact

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$$p(x,y) = p(a,b) = \frac{1}{12}.$$

Exercise 1 (a): Joint Entropy

$\log_2 12 = \log_2(2\times 2\times 3)$
$= 2 + \log_2 3$

Joint entropy

$$H(X,Y) = -\sum_{x,y} p_{X,Y}(x,y)\log_2 p_{X,Y}(x,y)$$
$$= -12\cdot\frac{1}{12}\log_2\frac{1}{12}$$
$$= 2+\log_2 3$$

Key fact

$$(a,b) \leftrightarrow (x,y)$$
$$p(x,y) = p(a,b) = \frac{1}{12}.$$

$\sqrt{\frac{10}{2}} = \sqrt{5}$

Exercise 1 (a): Joint Entropy

Joint entropy

$$
\begin{aligned}
H(X, Y) &= -\sum_{x,y} p_{X,Y}(x, y) \log_2 p_{X,Y}(x, y) \\
&= -12 \cdot \frac{1}{12} \log_2 \frac{1}{12} \\
&= 2 + \log_2 3
\end{aligned}
$$

Key fact

$$(a, b) \leftrightarrow (x, y)$$

$$p(x, y) = p(a, b) = \frac{1}{12}.$$

Exercise 1(a): $H(X)$

- Many (a, b) map to one x.
  e.g. $x{=}2$: $(2, 0), (1, 1)$.
- Each pair: $p_A(a)\, p_B(b) = \frac{1}{12}$.
- Group by $x{=}a{+}b$:

$$p_X(x) = \frac{1}{12}\, |\{(a, b) : a{+}b{=}x\}|.$$

- Plug into

$$H(X) = -\sum_x p_X(x) \log_2 p_X(x).$$

x	Events (a, b)	$p_X(x)$
1	$(1, 0)$	$1/12$
2	$(2, 0), (1, 1)$	$1/6$
3	$(3, 0), (2, 1)$	$1/6$
4	$(4, 0), (3, 1)$	$1/6$
5	$(5, 0), (4, 1)$	$1/6$
6	$(6, 0), (5, 1)$	$1/6$
7	$(6, 1)$	$1/12$

$$H(X) = -2 \cdot \frac{1}{12} \log_2 \frac{1}{12} - 5 \cdot \frac{1}{6} \log_2 \frac{1}{6}$$
$$= \frac{7}{6} + \log_2 3.$$

Exercise 1(a): $H(Y)$

- Many (a, b) map to one y. *e.g.* $y{=}1$: $(1, 0), (2, 1)$.
- Each pair: $p_A(a)\, p_B(b) = \frac{1}{12}$.
- Group by $y{=}a{-}b$:

$$p_Y(y) = \tfrac{1}{12}\, \left|\{(a, b) : a{-}b{=}y\}\right|.$$

- Plug into

$$H(Y) = -\sum_y p_Y(y) \log_2 p_Y(y).$$

y	Events (a, b)	$p_Y(y)$
0	$(1, 1)$	$1/12$
1	$(1, 0), (2, 1)$	$1/6$
2	$(2, 0), (3, 1)$	$1/6$
3	$(3, 0), (4, 1)$	$1/6$
4	$(4, 0), (5, 1)$	$1/6$
5	$(5, 0), (6, 1)$	$1/6$
6	$(6, 0)$	$1/12$

$$\begin{aligned} H(Y) &= -2 \cdot \tfrac{1}{12} \log_2 \tfrac{1}{12} - 5 \cdot \tfrac{1}{6} \log_2 \tfrac{1}{6} \\ &= \tfrac{7}{6} + \log_2 3. \end{aligned}$$

Exercise 1(a): $H(X|Y)$, $H(Y|X)$, $I(X;Y)$

Recall

- Chain rule:

$$
\begin{aligned}
H(X,Y) &= H(X) + H(Y|X) \\
&= H(Y) + H(X|Y)
\end{aligned}
$$

- Mutual information:

$$I(X;Y) = H(X) - H(X|Y)$$

Plug in $H(X,Y) = 2 + \log_2 3$ and $H(X) = H(Y) = \frac{7}{6} + \log_2 3$:

$$
\begin{aligned}
H(X|Y) &= H(X,Y) - H(Y) = \tfrac{5}{6} \\
H(Y|X) &= H(X,Y) - H(X) = \tfrac{5}{6} \\
I(X;Y) &= H(X) - H(X|Y) = \tfrac{1}{3} + \log_2 3
\end{aligned}
$$

Exercise 1 (b)

Question. For independent discrete random variables X and Y, show

$$I(X; X+Y) - I(Y; X+Y) = H(X) - H(Y).$$

$\boxed{X}$ $\boxed{X+Y}$

$\hookrightarrow Z = X+Y$

Exercise 1 (b)

$$Z = X + Y$$

Question. For independent discrete random variables X and Y, show

$$I(X; X+Y) - I(Y; X+Y) = H(X) - H(Y).$$

Solution.

$$I(X; X+Y) - I(Y; X+Y) = \boxed{\cancel{H(X+Y)} - H(X+Y|X)} - \cancel{H(X+Y)} + H(X+Y|Y).$$

$$I(X; Z) = H(Z) - H(Z|X)$$
$$= H(X+Y) - H(X+Y|X)$$

Exercise 1 (b)

Question. For independent discrete random variables X and Y, show
$$I(X;X+Y) - I(Y;X+Y) = H(X) - H(Y).$$

Solution.
$$\begin{aligned} I(X;X+Y) - I(Y;X+Y) &= H(X+Y) - H(X+Y|X) - H(X+Y) + H(X+Y|Y) \\ &= \boxed{H(X+Y|Y) - H(X+Y|X).} \end{aligned}$$

$$= H(X|Y) - H(Y|X)$$

Exercise 1 (b)

Question. For independent discrete random variables X and Y, show

$$I(X; X+Y) - I(Y; X+Y) = H(X) - H(Y).$$

Solution.

$$\begin{aligned} I(X; X+Y) - I(Y; X+Y) &= H(X+Y) - H(X+Y|X) - H(X+Y) + H(X+Y|Y) \\ &= H(X+Y|Y) - H(X+Y|X). \end{aligned}$$

Conditional entropy step

$$p(x+y|x) = p(y|x)$$

Given x, knowing $x+y$ is the same as knowing y:

$$H(X+Y|X) = H(Y|X).$$

Similarly,

$$H(X+Y|Y) = H(X|Y).$$

Exercise 1 (b)

Question. For independent discrete random variables X and Y, show

$$I(X; X+Y) - I(Y; X+Y) = H(X) - H(Y).$$

Solution.

$$\begin{aligned} I(X; X+Y) - I(Y; X+Y) &= H(X+Y) - H(X+Y|X) - H(X+Y) + H(X+Y|Y) \\ &= H(X+Y|Y) - H(X+Y|X). \end{aligned}$$

Conditional entropy step

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Given x, knowing $x+y$ is the same as knowing y:

$$H(X+Y|X) = H(Y|X).$$

Similarly,

$$H(X+Y|Y) = H(X|Y).$$

Remaining calculation

$$H(X+Y|Y) - H(X+Y|X)$$

Exercise 1 (b) $p(x|y) = \frac{p(x,y)}{p(y)}$ $\quad - \sum p(x|y) \log p(x|y) - \ldots$

Question. For independent discrete random variables X and Y, show

$$I(X; X+Y) - I(Y; X+Y) = H(X) - H(Y).$$

Solution.

$$\begin{aligned} I(X; X+Y) - I(Y; X+Y) &= H(X+Y) - H(X+Y|X) - H(X+Y) + H(X+Y|Y) \\ &= H(X+Y|Y) - H(X+Y|X). \end{aligned}$$

Conditional entropy step

$$p(x+y|x) = p(y|x)$$

Given x, knowing $x+y$ is the same as knowing y:

$$H(X+Y|X) = H(Y|X).$$

Similarly,

$$H(X+Y|Y) = H(X|Y).$$

Remaining calculation

$$\begin{aligned} &H(X+Y|Y) - H(X+Y|X) \\ &= H(X|Y) - H(Y|X) \end{aligned}$$

$$= H(X,Y) - H(Y)$$

$$\cancel{= H(X,Y)} + H(X)$$

$$= H(X) - H(Y)$$

Exercise 1 (b)

Question. For independent discrete random variables X and Y, show

$$I(X; X+Y) - I(Y; X+Y) = H(X) - H(Y).$$

Solution.

$$\begin{aligned} I(X; X+Y) - I(Y; X+Y) &= H(X+Y) - H(X+Y|X) - H(X+Y) + H(X+Y|Y) \\ &= H(X+Y|Y) - H(X+Y|X). \end{aligned}$$

Conditional entropy step

$$p(x+y|x) = p(y|x)$$

Given x, knowing $x+y$ is the same as knowing y:

$$H(X+Y|X) = H(Y|X).$$

Similarly,

$$H(X+Y|Y) = H(X|Y).$$

Remaining calculation

$$\begin{aligned} &H(X+Y|Y) - H(X+Y|X) \\ &= H(X|Y) - H(Y|X) \\ &= H(X) - H(Y), \end{aligned}$$

because X and Y are independent.

Exercise 2 (a)

$|Z$

Let X, Y and Z be three discrete random variables. The mutual information of X, Y and Z is defined as:

$$I(X;Y;Z) = \sum_x \sum_y \sum_z p(x,y,z) \log \left(\frac{p(x,y)p(x,z)p(y,z)}{p(x)p(y)p(z)p(x,y,z)} \right)$$

(a) Prove the lemma: $I(X;Y;Z) = I(X;Y) - I(X;Y|Z)$. Note that the conditional mutual information is defined as:

$$I(X;Y|Z) = \sum_x \sum_y \sum_z p(z)p(x,y|z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)}$$

Hint: Starting from the right hand side of the equation in the lemma.

Exercise 2 (a): Proof of $I(X;Y;Z) = I(X;Y) - I(X;Y|Z)$

$$\sum_x \sum_y p(x,y)\, Q(x,y)$$

$$I(X;Y) - I(X;Y|Z) = \underbrace{\sum_x \sum_y p(x,y) \log \frac{p(x,y)}{p(x)p(y)}} - \underbrace{\sum_z \sum_x \sum_y p(z)p(x,y|z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)}}$$

$$p(x,y) = \int_z p(x,y,z)\, dz$$

Exercise 2 (a): Proof of $I(X;Y;Z) = I(X;Y) - I(X;Y|Z)$

$$p(x,y|z) = \frac{p(x,y,z)}{p(z)}$$

$$\begin{aligned} I(X;Y) - I(X;Y|Z) &= \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{p(x)p(y)} - \sum_z \sum_x \sum_y p(z)p(x,y|z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)} \\ &= \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y)}{p(x)p(y)} - \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y|z)p(z)^2}{p(x|z)p(y|z)p(z)^2} \end{aligned}$$

Exercise 2 (a): Proof of $I(X;Y;Z) = I(X;Y) - I(X;Y|Z)$

$$
\begin{aligned}
I(X;Y) - I(X;Y|Z) &= \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{p(x)p(y)} - \sum_z \sum_x \sum_y p(z)p(x,y|z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)} \\
&= \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y)}{p(x)p(y)} - \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y|z)p(z)^2}{p(x|z)p(y|z)p(z)^2} \\
&= \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y)}{p(x)p(y)} - \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y,z)p(z)}{p(x,z)p(y,z)}
\end{aligned}
$$

$$E_{X,Y,Z}\left[\log \frac{p(X,Y)}{p(X)p(Y)} - \log \frac{p(X,Y,Z)\,p(Z)}{p(X,Z)\,p(Y,Z)}\right]$$

Exercise 2 (a): Proof of $I(X;Y;Z) = I(X;Y) - I(X;Y|Z)$

$$
\begin{aligned}
I(X;Y) - I(X;Y|Z) &= \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{p(x)p(y)} - \sum_z \sum_x \sum_y p(z)p(x,y|z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)} \\
&= \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y)}{p(x)p(y)} - \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y|z)p(z)^2}{p(x|z)p(y|z)p(z)^2} \\
&= \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y)}{p(x)p(y)} - \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y,z)p(z)}{p(x,z)p(y,z)} \\
&= \boxed{\sum_{x,y,z} p(x,y,z) \log \left(\frac{p(x,y)p(x,z)p(y,z)}{p(x)p(y)p(z)p(x,y,z)} \right)}
\end{aligned}
$$

Exercise 2 (a): Proof of $I(X;Y;Z) = I(X;Y) - I(X;Y|Z)$

$$
\begin{aligned}
I(X;Y) - I(X;Y|Z) &= \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{p(x)p(y)} - \sum_z \sum_x \sum_y p(z)p(x,y|z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)} \\
&= \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y)}{p(x)p(y)} - \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y|z)p(z)^2}{p(x|z)p(y|z)p(z)^2} \\
&= \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y)}{p(x)p(y)} - \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y,z)p(z)}{p(x,z)p(y,z)} \\
&= \sum_{x,y,z} p(x,y,z) \log \left(\frac{p(x,y)p(x,z)p(y,z)}{p(x)p(y)p(z)p(x,y,z)} \right) \\
&= I(X;Y;Z)
\end{aligned}
$$

Exercise 2 (b)

$$\boxed{I(X;Y)} - \cancel{I(X;Y;Z)} + \cancel{I(Y;Z)}$$
$$- \cancel{I(Y;Z)} + \cancel{I(X;Y;Z)}$$

Question. Prove

$$I(X;Y) = I(X;Y|Z) + I(Y;Z) - I(Y;Z|X).$$

Lemma

$$I(X;Y;Z) = I(X;Y) - I(X;Y|Z).$$

Computation. Apply the lemma twice:

$$I(X;Y|Z) = I(X;Y) - I(X;Y;Z), \quad I(Y;Z|X) = I(Y;Z) - I(X;Y;Z).$$

Then

Exercise 2 (b)

Question. Prove

$$I(X;Y) = I(X;Y|Z) + I(Y;Z) - I(Y;Z|X).$$

Lemma

$$I(X;Y;Z) = I(X;Y) - I(X;Y|Z).$$

Computation. Apply the lemma twice:

$$I(X;Y|Z) = I(X;Y) - I(X;Y;Z), \quad I(Y;Z|X) = I(Y;Z) - I(X;Y;Z).$$

Then

Exercise 2 (b)

Question. Prove

$$I(X; Y) = I(X; Y|Z) + I(Y; Z) - I(Y; Z|X).$$

Lemma

$$I(X; Y; Z) = I(X; Y) - I(X; Y|Z).$$

Computation. Apply the lemma twice:

$$I(X; Y|Z) = I(X; Y) - I(X; Y; Z), \quad I(Y; Z|X) = I(Y; Z) - I(X; Y; Z).$$

Then

$$I(X; Y|Z) + I(Y; Z) - I(Y; Z|X)$$

Exercise 2 (b)

Question. Prove

$$I(X;Y) = I(X;Y|Z) + I(Y;Z) - I(Y;Z|X).$$

Lemma

$$I(X;Y;Z) = I(X;Y) - I(X;Y|Z).$$

Computation. Apply the lemma twice:

$$I(X;Y|Z) = I(X;Y) - I(X;Y;Z), \quad I(Y;Z|X) = I(Y;Z) - I(X;Y;Z).$$

Then

$$\begin{aligned} &I(X;Y|Z) + I(Y;Z) - I(Y;Z|X) \\ &= (I(X;Y) - I(X;Y;Z)) + I(Y;Z) - (I(Y;Z) - I(X;Y;Z)) \end{aligned}$$

Exercise 2 (b)

Question. Prove

$$I(X;Y) = I(X;Y|Z) + I(Y;Z) - I(Y;Z|X).$$

Lemma

$$I(X;Y;Z) = I(X;Y) - I(X;Y|Z).$$

Computation. Apply the lemma twice:

$$I(X;Y|Z) = I(X;Y) - I(X;Y;Z), \quad I(Y;Z|X) = I(Y;Z) - I(X;Y;Z).$$

Then

$$\begin{aligned}
&I(X;Y|Z) + I(Y;Z) - I(Y;Z|X) \\
&= (I(X;Y) - I(X;Y;Z)) + I(Y;Z) - (I(Y;Z) - I(X;Y;Z)) \\
&= I(X;Y).
\end{aligned}$$

Exercise 3: Smoothed Cross-Entropy Loss

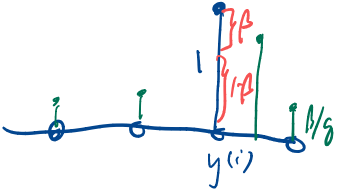
$\sum_{i} \log P(y^{(i)})$

Over-confidence: model is more confident in its prediction than the data warrants. **Label smoothing** (Szegedy et al., 2016) alleviates this and improves robustness.
Conventional cross-entropy minimizes the KL-divergence between the ground-truth distribution d (a delta: $d_k = 1$ for the true class) and the predicted distribution $\pi(\mathbf{x}|\boldsymbol{\theta})$.
Label smoothing smooths d: given β (e.g. $\beta = 0.1$), distribute probability mass β uniformly across the g classes and reduce the ground-truth mass:

$\pi(x|\theta)$

$P(y^{(i)})$

$$\tilde{d}_k = \begin{cases} \beta/g & \text{for } d_k = 0, \\ 1 - \beta + \beta/g & \text{for } d_k = 1. \end{cases}$$



The smoothed cross-entropy is then $D_{KL}(\tilde{d} \,\|\, \pi(\mathbf{x}|\boldsymbol{\theta}))$.

(a) Derive the empirical risk when using the smoothed cross-entropy as loss function. *Hint: some terms can be merged into a constant and ignored during implementation.*

Exercise 3 (a): $D_{KL}(\tilde{d} \,\|\, \pi(\mathbf{x}|\boldsymbol{\theta}))$

Start from the KL definition and expand:

$$
\begin{aligned}
R_{\text{emp}}(\boldsymbol{\theta}) &= \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{g} \tilde{d}_k^{(i)} \log \frac{\tilde{d}_k^{(i)}}{\pi_k(\mathbf{x}^{(i)}|\boldsymbol{\theta})} \\
&= \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{g} \Big(\underbrace{\tilde{d}_k^{(i)} \log \tilde{d}_k^{(i)}}_{\text{indep. of } \boldsymbol{\theta}} - \tilde{d}_k^{(i)} \log \pi_k(\mathbf{x}^{(i)}|\boldsymbol{\theta}) \Big).
\end{aligned}
$$

Recall: $\tilde{d}_k$

$$
\tilde{d}_k = \begin{cases} \beta/g, & d_k = 0 \\ 1-\beta+\beta/g, & d_k = 1 \end{cases}
$$

Drop the first term (constant w.r.t. θ) and plug in $\tilde{d}_k$ (with $y^{(i)}$ the true class) to get the practical form:

$$
R_{\text{emp}}(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[(1-\beta)\log \pi_{y^{(i)}}(\mathbf{x}^{(i)}|\theta) + \frac{\beta}{g}\sum_{k=1}^{g}\log \pi_k(\mathbf{x}^{(i)}|\theta)\right] + \text{const.}
$$

Exercise 3 (a): $D_{KL}(\tilde{d} \,\|\, \pi(\mathbf{x}|\boldsymbol{\theta}))$

Start from the KL definition and expand:

$$
\begin{aligned}
R_{\text{emp}}(\boldsymbol{\theta}) &= \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{g} \tilde{d}_k^{(i)} \log \frac{\tilde{d}_k^{(i)}}{\pi_k(\mathbf{x}^{(i)}|\boldsymbol{\theta})} \\
&= \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{g} \Big(\underbrace{\tilde{d}_k^{(i)} \log \tilde{d}_k^{(i)}}_{\text{indep. of } \boldsymbol{\theta}} - \tilde{d}_k^{(i)} \log \pi_k(\mathbf{x}^{(i)}|\boldsymbol{\theta}) \Big).
\end{aligned}
$$

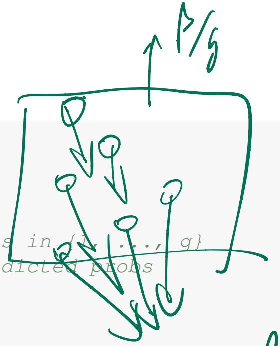
Recall: $\tilde{d}_k$

$$
\tilde{d}_k = \begin{cases} \beta/g, & d_k = 0 \\ 1-\beta+\beta/g, & d_k = 1 \end{cases}
$$

Drop the first term (constant w.r.t. $\boldsymbol{\theta}$) and plug in $\tilde{d}_k$ (with $y^{(i)}$ the true class) to get the practical form:

$$
R_{\text{emp}}(\boldsymbol{\theta}) = -\frac{1}{n}\sum_{i=1}^{n}\left[(1-\beta)\log \pi_{y^{(i)}}(\mathbf{x}^{(i)}|\boldsymbol{\theta}) + \frac{\beta}{g}\sum_{k=1}^{g}\log \pi_k(\mathbf{x}^{(i)}|\boldsymbol{\theta})\right] + \text{const.}
$$

Exercise 3 (b): Implementation (Python)

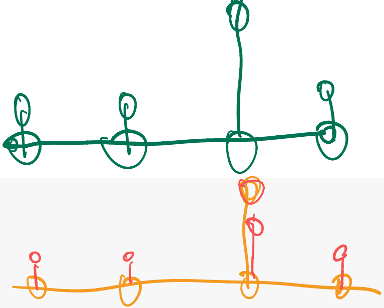


(b) Implement the smoothed cross-entropy. Vectorised `numpy` version:

```
import numpy as np
from numpy.typing import NDArray


def smoothed_ce_loss(label: NDArray,      # shape (n,), values in {1, ..., g}
                     pred: NDArray,       # shape (n, g), predicted probs
                     smoothing: float) -> float:
    assert 0.0 <= smoothing <= 1.0
    n, g = pred.shape
    assert label.shape == (n,)

    # Build smoothed label matrix: start uniform, set true class to 1 - b + b/g
    base = smoothing / g
    smoothed = np.full((n, g), base)
    smoothed[np.arange(n), label - 1] = 1.0 - smoothing + base

    # Per-sample cross-entropy, then mean over the batch
    return float(-np.sum(smoothed * np.log(pred), axis=1).mean())
```

$1 - \beta + \beta/g$

Exercise 3 (b): Demo Runs



A “confident” toy model: 5 samples, 3 classes.

```
label = np.array([1, 2, 2, 3, 1])
pred  = np.array([[0.85, 0.10, 0.05],
                  [0.05, 0.90, 0.05],
                  [0.02, 0.95, 0.03],
                  [0.13, 0.02, 0.85],
                  [0.86, 0.04, 0.10]])

smoothed_ce_loss(label, pred, smoothing=0.0)    # plain CE        -> 0.1265
smoothed_ce_loss(label, pred, smoothing=0.2)    # label-smoothed  -> 0.5121
```

Takeaway

- Loss jumps $\sim 4\times$ (0.13 $\to$ 0.51) with $\beta = 0.2$.
- Target is no longer a delta $\Rightarrow$ a confident peak on the true class is now *penalised*.

Summary — Connecting the exercise to the lecture

	What we derived	Lecture concept made concrete
1 (a)	$H(X), H(Y\|X), H(X,Y), I(X;Y)$ from the joint table; identity $H(X\|Y) = H(X,Y) - H(Y)$.	Entropy + conditional/joint entropy; MI as $H(X) - H(X\|Y)$.
1 (b)	MI identity $I(X; X+Y) - I(Y; X+Y) = H(X) - H(Y)$ via conditional MI manipulation.	Symmetry and chain properties of (conditional) MI.
2	Lemma $I(X;Y;Z) = I(X;Y) - I(X;Y\|Z)$; corollary $I(X;Y) = I(X;Y\|Z) + I(Y;Z) - I(Y;Z\|X)$.	Three-variable MI: chain rule and inclusion-exclusion structure.
3	$R_{\text{emp}} = -\frac{1}{n}\sum_i \left[(1-\beta)\log \pi_{y^{(i)}} + \frac{\beta}{g}\sum_k \log \pi_k\right]$ + vectorised `numpy` loss.	Cross-entropy as a KL objective; entropy of target is a $\boldsymbol{\theta}$-free constant; label smoothing as a calibration tool.